PAKISTANI FASHION INTELLIGENCE

Fashion Brand Competitive Price Analytics

A Data Driven Comparison of Leading Pakistani Fashion Brands

Prepared By	Aqib Hanif
Assigned By	Mam Sumayyea Salahuddin
Institute	Arfa Karim Incubation Center, Peshawar
Project Region	Peshawar, Pakistan
Brands	J. Maria.B Sana Safinaz
Project Date	31 August 2026

Project Report

Project Overview

I developed this project to compare the pricing and promotional behavior of three Pakistani fashion brands using one consistent analytical process. The brands included in my analysis are J., Maria.B, and Sana Safinaz. I collected product level information from the official online stores, prepared the data in Python, analyzed it in PostgreSQL, added Peshawar weather context through an API, and presented the results in an interactive Excel dashboard.

Business Problem

Fashion brands may sell similar categories, but their price levels, discount activity, product mix, and market position can be very different. The problem I addressed was how to compare these brands with the same measures so that their pricing strategy and competitive position become easy to understand.

Project Objectives

1. Collect a practical product sample from the official websites of J., Maria.B, and Sana Safinaz.
2. Clean the raw data and create fields that support pricing and discount analysis.
3. Use PostgreSQL to answer business questions with structured SQL queries.
4. Use the Open Meteo API to add Peshawar weather and seasonal context.
5. Build an interactive Excel dashboard with clear filters, KPI cards, and business focused charts.
6. Present findings that can be explained easily in a classroom presentation.

Scope

The project contains 281 products across five main categories. Product prices come from the brands online stores. Peshawar is used for the weather API, seasonal context, and presentation region. I do not treat Peshawar weather as a direct cause of fashion prices.

Project at a Glance

Measure	Result
Brands	3
Products	281
Categories	5
Overall Median Price	PKR 12,990
Overall Average Discount	4.62%
Products on Sale	13.88%

Tools and Technologies

Tool or Technology	How I Used It
Python and Jupyter Notebook	Web scraping, data cleaning, feature engineering, and API work
Requests and BeautifulSoup	Opening collection pages and extracting product links and

	product information
Pandas	Cleaning, checking, transforming, grouping, and exporting the dataset
PostgreSQL	Database storage, SQL analysis, ranking, aggregation, and dashboard preparation
Open Meteo API	Current Peshawar weather and seasonal weather context
Microsoft Excel	Interactive filters, KPI cards, charts, and final dashboard
CSV Files	Moving data between Python, PostgreSQL, and Excel

Data Collection with Web Scraping

I used my Jupyter web scraping notebook to collect product records from the official websites of the three selected brands. My target was approximately 100 products per brand, while keeping the sample spread across useful clothing categories.

Brand	Products Collected
J.	100
Maria.B	100
Sana Safinaz	81
Total	281

The raw dataset includes product identification, brand, product name, SKU, category, subcategory, collection, original price, sale price, discount amount, discount percentage, availability, currency, product URL, source website, and scraping date.

I used one request session, browser style headers, page level product link collection, duplicate checks, and a small delay between requests. I first tried the Shopify product JSON endpoint and used structured product data as a fallback when needed.

Data Cleaning and Feature Engineering

The raw file contained 281 records and the processed file also contained 281 records. The source snapshot did not contain duplicate product URLs or missing values in the main product fields. I still performed cleaning steps so the same rules were applied to every brand.

1. Removed extra spaces from text columns and standardized important category names.
2. Converted date fields into date values and price fields into numeric values.
3. Checked that sale price was not higher than original price.
4. Recalculated discount amount and discount percentage from cleaned prices.
5. Created sale status, price segment, discount band, category median price, and price index.
6. Created brand level scoring fields for price competitiveness, discount strategy, product variety, and availability.

Price Segments

Segment	Rule
Budget	Below PKR 8,000
Mid Range	PKR 8,000 to PKR 14,999
Premium	PKR 15,000 to PKR 29,999
Luxury	PKR 30,000 and above

Brand Intelligence Score

I created one score out of 100 to combine important brand measures. The score gives 30% weight to price competitiveness, 25% to discount strategy, 25% to product variety, and 20% to availability.

Brand	Brand Intelligence Score
Sana Safinaz	81.18
J.	65.28
Maria.B	45.32

PostgreSQL Analysis

I created a PostgreSQL database named fashion_analytics and a main table named fashion_products. The processed CSV was imported into this table. I used data type rules and simple constraints to prevent invalid price values.

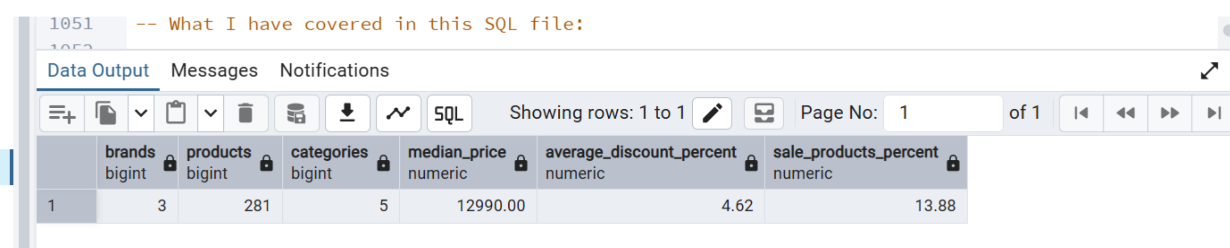
My SQL work covers SELECT, WHERE, GROUP BY, ORDER BY, CASE WHEN, aggregate functions, PERCENTILE_CONT for median price, common table expressions, ROW_NUMBER, RANK, window functions, a reusable view, and indexes.

Main SQL Questions

1. Which brand has the lowest median price?
2. Which brand has the highest average discount?
3. What percentage of products are on sale for each brand?
4. Which categories contain the most products?
5. How does product mix differ by brand and category?
6. How are products distributed across Budget, Mid Range, Premium, and Luxury price segments?
7. Which products have the highest savings?
8. How do the brands rank using the Brand Intelligence Score?

```sql
-- STEP 31 - FINAL PROJECT RESULT
-- ============================================================

-- This is the final overall summary I can show in class.

SELECT
    COUNT(DISTINCT brand) AS brands,

    COUNT(*) AS products,

    COUNT(DISTINCT category) AS categories,

    ROUND(
        PERCENTILE_CONT(0.5)
        WITHIN GROUP (
            ORDER BY sale_price
        )::NUMERIC,
        2
    ) AS median_price,

    ROUND(
        AVG(discount_percent),
        2
    ) AS average_discount_percent,

    ROUND(
        100.0 *
        SUM(
            CASE
                WHEN sale_status = 'On Sale' THEN 1
                ELSE 0
            END
        )
        / COUNT(*),
        2
    ) AS sale_products_percent

FROM fashion_products;



-- ============================================================
```

Figure 1. Final PostgreSQL project result query from my SQL file



	brands bigint	products bigint	categories bigint	median_price numeric	average_discount_percent numeric	sale_products_percent numeric
1	3	281	5	12990.00	4.62	13.88

Figure 2. PostgreSQL final project result in pgAdmin

The PostgreSQL output confirmed 3 brands, 281 products, 5 categories, an overall median price of PKR 12,990, an average discount of 4.62%, and 13.88% of products on sale.

Peshawar Weather API

I used the Open Meteo API without an API key. The purpose of this step was to add regional and seasonal context for Peshawar. The current weather file and the six month seasonal file were exported as CSV files for use in Excel.

Weather Measure	Value on 31 August 2026
Temperature	33.1 C
Condition	Clear
Maximum Temperature	34.0 C
Minimum Temperature	25.9 C
Humidity	62%
Precipitation	0.0 mm
Wind Speed	5.0 km/h

For seasonal context, I prepared monthly values for February 2026 to July 2026. The table contains average maximum temperature, average minimum temperature, and total precipitation. This weather information is contextual and is not used to claim that weather caused the product prices.

Interactive Excel Dashboard

I created the final dashboard in Microsoft Excel so the project can be presented without Power BI or Tableau. The dashboard uses dropdown filters for Brand, Category, Subcategory, Collection, Availability, and Price Segment. The filters update the KPI values and chart calculations.

Final Excel Dashboard

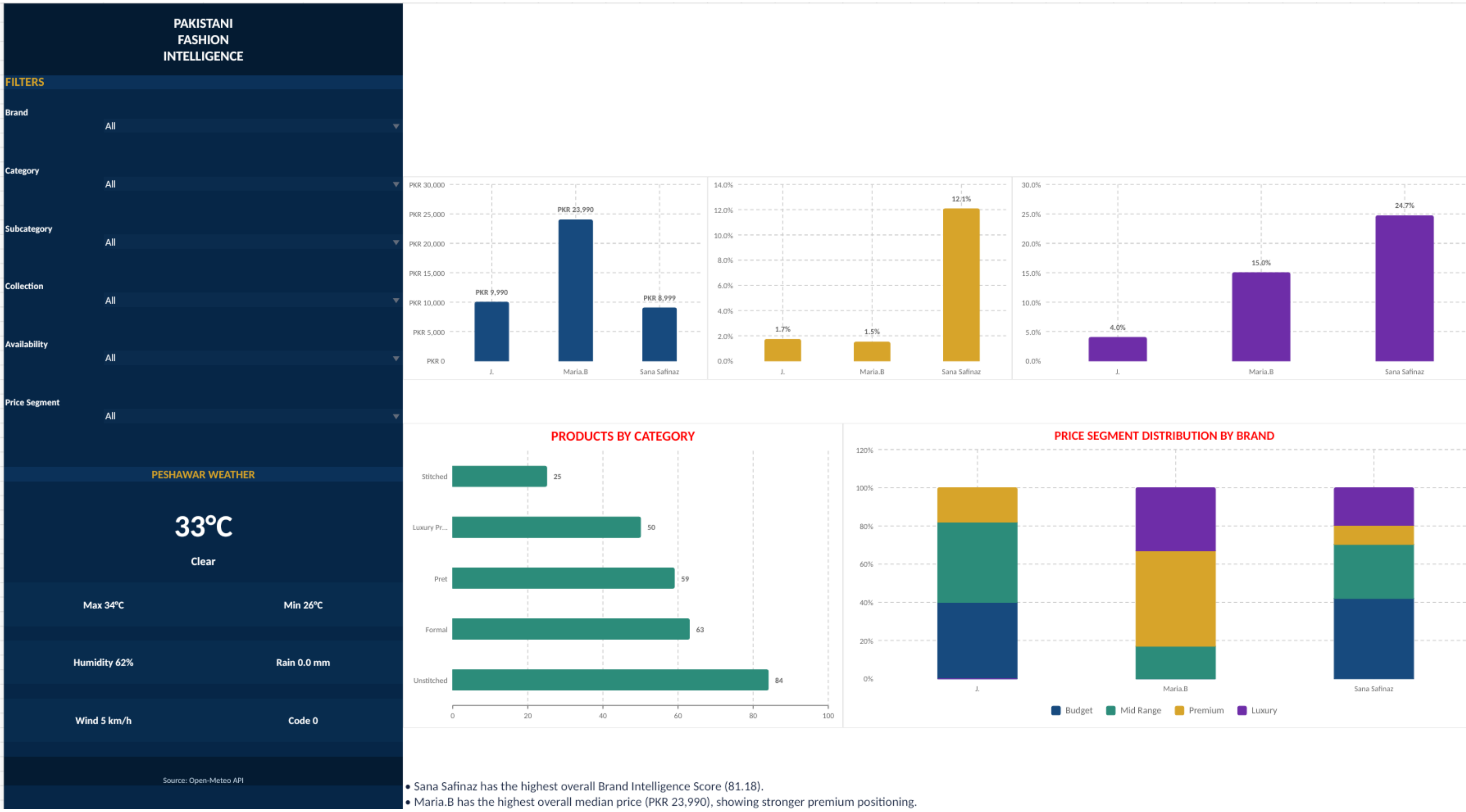


Figure 3. My final interactive Excel dashboard

Findings

Finding	Result
Overall market snapshot	281 products, 5 categories, median price PKR 12,990
Highest median price	Maria.B at PKR 23,990
J. median price	PKR 9,990
Sana Safinaz median price	PKR 8,999
Highest average discount	Sana Safinaz at 12.07%
J. average discount	1.70%
Maria.B average discount	1.50%
Highest share of sale products	Sana Safinaz at 24.69%
Maria.B sale products	15.00%
J. sale products	4.00%
Largest category	Unstitched with 84 products
Highest Brand Intelligence Score	Sana Safinaz at 81.18

Price Segment Positioning

J. is strongly concentrated in Budget and Mid Range products. 40% of its products are Budget and 42% are Mid Range. Maria.B has a much more premium structure, with 50% Premium products and 33% Luxury products. Sana Safinaz has a mixed position, with 42.0% Budget, 28.4% Mid Range, 9.9% Premium, and 19.8% Luxury products.

Category Mix

Across all brands, Unstitched is the largest category with 84 products. Formal contains 63 products, Pret contains 59, Luxury Pret contains 50, and Stitched contains 25.

Business Recommendations

1. For J., I would protect the affordable price position and use targeted promotions on higher priced products rather than discounting the whole range.
2. For Maria.B, I would maintain the premium position but use selective promotional campaigns to improve conversion without reducing the premium image.
3. For Sana Safinaz, I would continue using discount activity carefully because the brand currently has the strongest promotion metrics and the highest Brand Intelligence Score.
4. For all three brands, I would repeat the scraping process on a regular schedule so that future analysis can measure real price movement over time.
5. I would also add sales volume or customer demand data in a future version because product prices alone cannot show which strategy produces the strongest sales performance.

Limitations and Future Improvement

1. The current product analysis is a single price snapshot, so it cannot prove long term price trends.
2. Website structure can change, which may require updates to the scraping logic.
3. All sampled products were marked available at the time of the current scraping process, so the availability measure does not strongly separate the brands.
4. The project does not include actual sales quantity, customer demographics, profit, or conversion data.
5. Weather is used only as Peshawar seasonal context and should not be interpreted as a direct reason for pricing changes.

Conclusion

This project gave me a complete data analyst workflow from data collection to final presentation. I used web scraping to collect real product information, Python and Pandas to prepare the dataset, PostgreSQL to answer business questions, the Open Meteo API to add Peshawar context, and Excel to build an interactive dashboard.

The analysis shows clear differences between the three brands. Maria.B has the highest median price and a strong Premium and Luxury position. Sana Safinaz has the strongest current discount activity and the highest Brand Intelligence Score. J. has a more affordable price structure and a large concentration in Budget and Mid Range products. These differences are easier to compare because I used the same fields, calculations, and dashboard structure for all three brands.

Project Files

File	Purpose
01_Fashion_Web_Scraping_Aqib_Hanif.ipynb	Web scraping and raw data collection
01_raw_fashion_products.csv	Raw product data
02_Data_Cleaning_Feature_Engineering_Aqib_Hanif.ipynb	Cleaning and feature engineering
02_processed_fashion_products.csv	Processed analytical dataset
03_PostgreSQL_Analysis_Aqib_Hanif.sql	Database creation and SQL analysis
04_Peshawar_Weather_API_Aqib_Hanif.ipynb	Open Meteo API integration
04_current_peshawar_weather.csv	Current Peshawar weather
04_peshawar_seasonal_context.csv	Seasonal Peshawar weather context
05_Pakistani_Fashion_Intelligence_Dashboard_Aqib_Hanif.xlsx	Final interactive Excel dashboard